A MATHEMATICAL MULTISCALE FRAMEWORK COUPLING WITHIN-HOST RESISTANCE EVOLUTION AND BETWEEN-HOST TRANSMISSION

Felipe Schardong, Claudio Jose Struchiner, Luiz Max Carvalho
School of Applied Mathematics, Getulio Vargas Foundation (FGV-EMAp)

Introduction

Antimicrobial resistance (AMR) is one of the leading global public health threats, causing more than 1.2 million attributable deaths in 2019 and projected to cause over 39 million cumulative deaths between 2025 and 2050 [1, 2]. Resistance emerges within hosts through mutation and selection under drug exposure, while its dissemination depends on transmission processes and host mobility [3, 4]. Although within-host resistance evolution and between-host transmission have been extensively studied [5, 6], they are often modeled separately [7]. Here, we develop a multiscale framework coupling pharmacokinetics and pharmacodynamics to stochastic within-host resistance evolution and a two-strain SEIRS model of between-host transmission. We apply this framework to investigate how treatment non-adherence affects resistance emergence and spread, also considering treatment abandonment and delayed initiation. The framework is first applied to a single host population and then extended to a two-patch metapopulation.

Methods

An overview of the general mathematical framework is illustrated in Figure 1, linking drug dosing to population-level spatial spread.

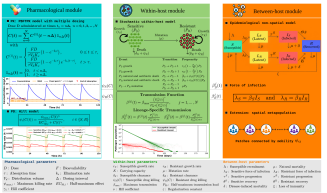


Figure 1: Schematic representation of the multiscale framework coupling within-host resistance evolution and between-host transmission. Source: Author.

For each scenario, we define non-adherence sub-levels, summarized in Table 1, to assess how deviations from the prescribed regimen affect resistance development.

Table 1: Treatment adherence scenarios and operational parameters.

Scenario	Description	Parameter levels
1. Intermittent non-adherence	Randomly missed doses throughout treatment	Low: 5%; Moderate: 25%; High: 50% missed doses
2. Treatment abandonment	Permanent treatment discontinuation without resumption	30, 60, and 90 days.
3. Delayed treatment initiation	Untreated infection period before starting treatment	Short: 15; Moderate: 30; Long: 60 days

Results

Figures 2 and 3 show the resistance dynamics under intermittent non-adherence, treatment abandonment, and delayed initiation. Figure 4 compares the three scenarios across cumulative mortality, resistant strain dominance, and outbreak peak size. Figure 5 compares resistance spread under symmetric vs. asymmetric migration ($\eta = 5$).

Scenario 1 – Intermittent non-adherence

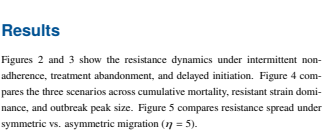


Figure 2: Adherence effects at 100%, 95%, 75%, and 50% on multiscale resistance dynamics. Panels a–d correspond to the four adherence rates. Source: Author.

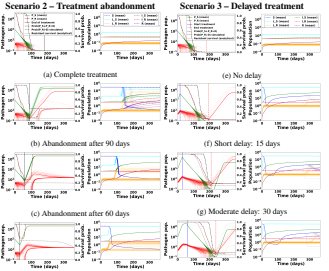


Figure 3: Resistance dynamics under treatment abandonment (a–d) and delayed initiation (e–h). Pathogen population (left axis) and host subpopulations (right axis). Source: Author.

Synthesis of treatment non-adherence scenarios

The effects of non-adherence are nonlinear. Small lapses have limited effects, whereas lower adherence or early treatment interruption can lead to outbreaks. Delayed treatment leads to the largest outbreaks.

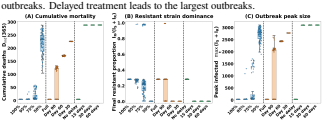
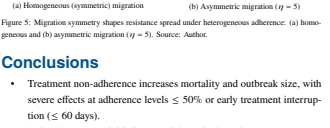


Figure 4: Comparative synthesis of treatment non-adherence scenarios. Source: Author.

Spatial spread: symmetric vs. asymmetric migration

Resistance rescue in Patch 2 is rare at low connectivity ($\nu \leq 10^{-1}$) but nearly certain from baseline connectivity onward ($\nu \geq 10^{0}$). Asymmetric infectious outflow ($\eta = 5$) lowers this migration threshold and amplifies the resistant burden at intermediate connectivity.



(a) Homogeneous (symmetric) migration (b) Asymmetric migration ($\eta = 5$)

Figure 5: Migration symmetry shapes resistance spread under heterogeneous adherence: (a) homogeneous and (b) asymmetric migration ($\eta = 5$). Source: Author.

Conclusions

- Treatment non-adherence increases mortality and outbreak size, with severe effects at adherence levels $\leq 50\%$ or early treatment interruption (≤ 60 days).
- Delayed treatment initiation (≥ 15 days) leads to the worst outcomes, with further delays having limited additional effects.
- The decline in the resistant fraction (Fig. 4B) reflects dilution by susceptible strain expansion rather than a reduction in the absolute resistant load.
- Asymmetric migration promotes resistant strain spread from non-adherent hotspots, enabling stochastic rescue and late outbreaks in adherent communities.
- Effective AMR control requires adherence support and targeted infection control in hotspots and surrounding communities.

References